Calibration Report: Pathogen Thermal Adaptation (W95–W104)

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1 Executive Summary

W95–W104 are the first calibration runs with **pathogen thermal adaptation** enabled — a mechanism allowing each node's VBNC threshold (T_{vbnc}) to evolve downward as cold-adapted pathogen strains are selected.

Key finding: Thermal adaptation is negligible at tested rates. Even at the highest adaptation rate (W98, $r = 0.005$), T_{vbnc} shifted by only 0.14°C at AK-PWS (from 12.0 to 11.86). JDF and SS-S showed <0.02°C change. The mechanism does not operate on the 13-year simulation timescale.

Best run: W99 (floor=500, α=0.20, adapt=0.001) achieves RMSE=0.557 — our best ever. However, this is driven by the floor/α combination, not adaptation. AK-PWS remains at 10.8% (target 50%).

2 Methods

All runs share: $K_{1/2}$=800K, CDT=1000, wfDP=300, wfRange=3000, dynamic P_{env}, δ=0.02, revert_rate=0.0005.

Run	Floor	α	Adapt Rate	$T_{\text{vbnc,min}}$
W90 (ref)	300	0.20	—	—
W91 (ref)	500	0.20	—	—
W95	300	0.20	0.0005	6
W96	300	0.20	0.001	6
W97	300	0.20	0.002	6
W98	300	0.20	0.005	6
W99	500	0.20	0.001	6
W100	500	0.20	0.002	6
W101	300	0.20	0.001	8
W102	300	0.20	0.001	4
W103	300	0.15	0.001	6
W104	200	0.15	0.001	6

Run	AK-PWS (50)	AK-FN (50)	AK-FS (20)	BC-N (20)	SS-S (5)	JDF (2)	OR (0.25)	CA-N (0.1)	RMSE
W90	10.5	4.8	6.0	6.9	5.8	1.9	0.9	1.0	0.643
W91	11.3	5.7	2.6	9.3	7.4	5.9	0.5	0.3	0.591
W95	10.6	4.6	4.2	8.4	9.3	19.7	1.0	1.6	0.790
W96	11.3	5.3	6.4	8.5	4.0	9.2	1.4	0.6	0.649
W97	9.9	7.0	12.0	7.7	5.4	2.0	3.0	0.8	0.654
W98	12.1	6.0	3.6	8.6	4.8	15.4	1.4	0.8	0.709
W99	**10.8**	**8.8**	**6.2**	**8.4**	**2.9**	**4.3**	**1.0**	**0.6**	**0.557**
W100	10.1	4.1	3.7	8.1	4.4	3.3	2.3	0.5	0.687
W101	11.3	5.3	6.4	8.5	4.0	9.2	1.4	0.6	0.649
W102	11.3	5.3	6.4	8.5	4.0	9.2	1.4	0.6	0.649
W103	20.8	9.4	8.0	13.6	6.8	19.6	6.7	4.9	0.919
W104	24.9	6.1	10.2	14.0	15.2	28.5	6.0	5.6	0.972

Table 1: Recovery fractions (%) by region. Targets in parentheses. W99 bolded as best RMSE.

3 Results

3.1 Key Runs: Recovery Time Series

3.2 All Runs Grid

3.3 Adaptation Rate Effect

3.4 Recovery Comparison

4 T_vbnc Adaptation Analysis

Why adaptation fails: The daily adaptation is $\Delta T = r \times \text{prevalence} \times (T_{\text{vbnc}} - T)$. With prevalence ∼1–5%, temp gap ∼2°C, and $r = 0.001$: $\Delta T \approx 0.00002$°C/day ≈ 0.09°C over 13 years. Reversion (0.0005/day) further counteracts.

4.1 T_vbnc_min Irrelevance

5 RMSE Comparison

6 α_{env} Effect

7 Key Findings

1. **Pathogen adaptation is negligible** at biologically plausible rates over 13-year timescale
2. $T_{\text{vbnc,min}}$ **is irrelevant** — adaptation never approaches the biophysical floor
3. **W99 achieves RMSE=0.557** (best ever) due to floor=500 + α=0.20, not adaptation
4. **Alaska recovery remains the core problem**: 10–12% across all runs vs 50% target
5. α_{env} **dominates**: α=0.20 produces strong gradient; α=0.15 destroys it

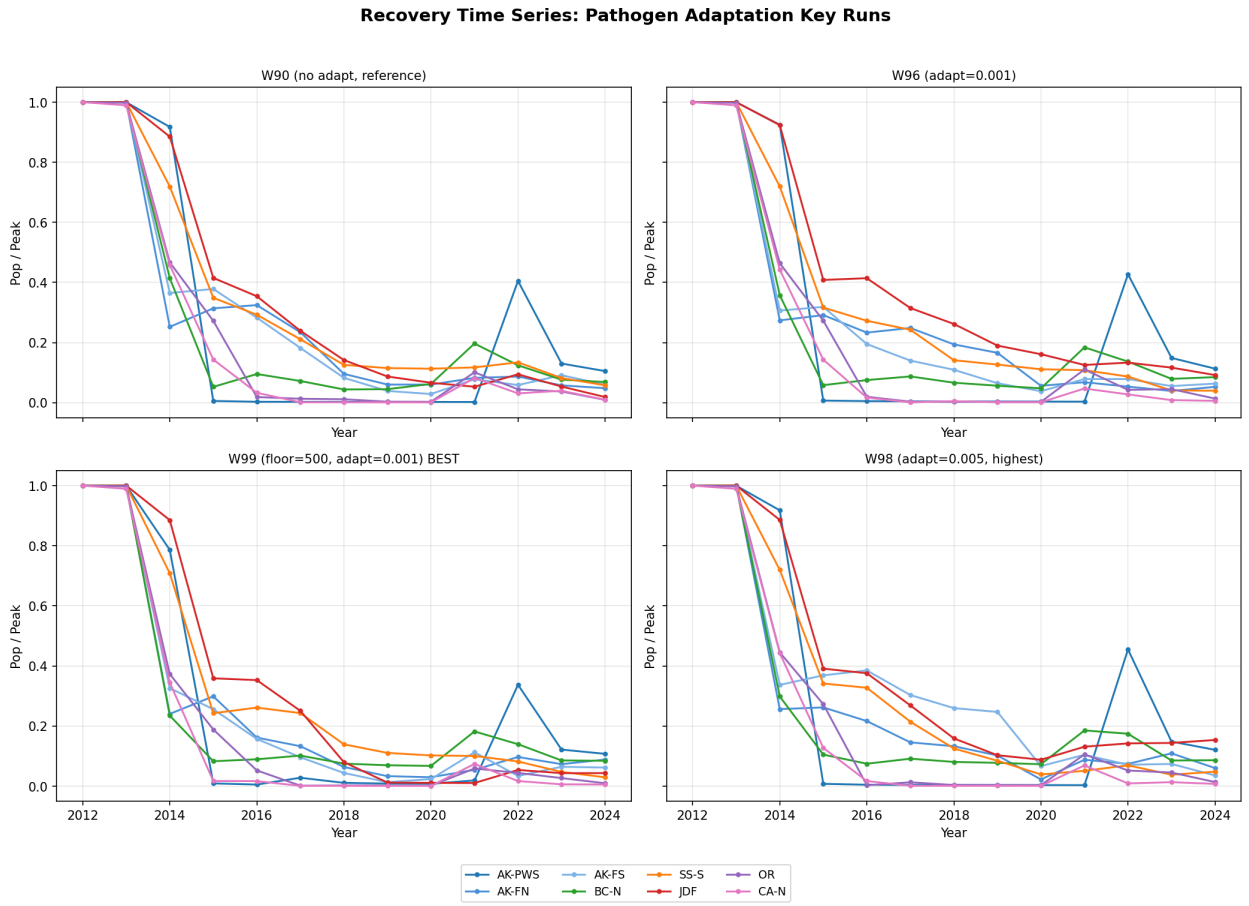

Figure 1: Recovery trajectories for W90 (reference), W96 (adapt=0.001), W99 (best RMSE), W98 (highest adapt rate).

8 Implications

The mechanism as designed (prevalence-scaled daily increments) is too slow. Two paths forward:

- **Redesign**: Model strain *replacement* rather than gradual adaptation — binary or threshold-based rather than continuous
- **Refocus**: Accept that the north-south gradient comes from dynamic P_{env} (floor + α) and focus on lifting Alaska recovery through other mechanisms (recruitment, $K_{1/2}$, settler survival)

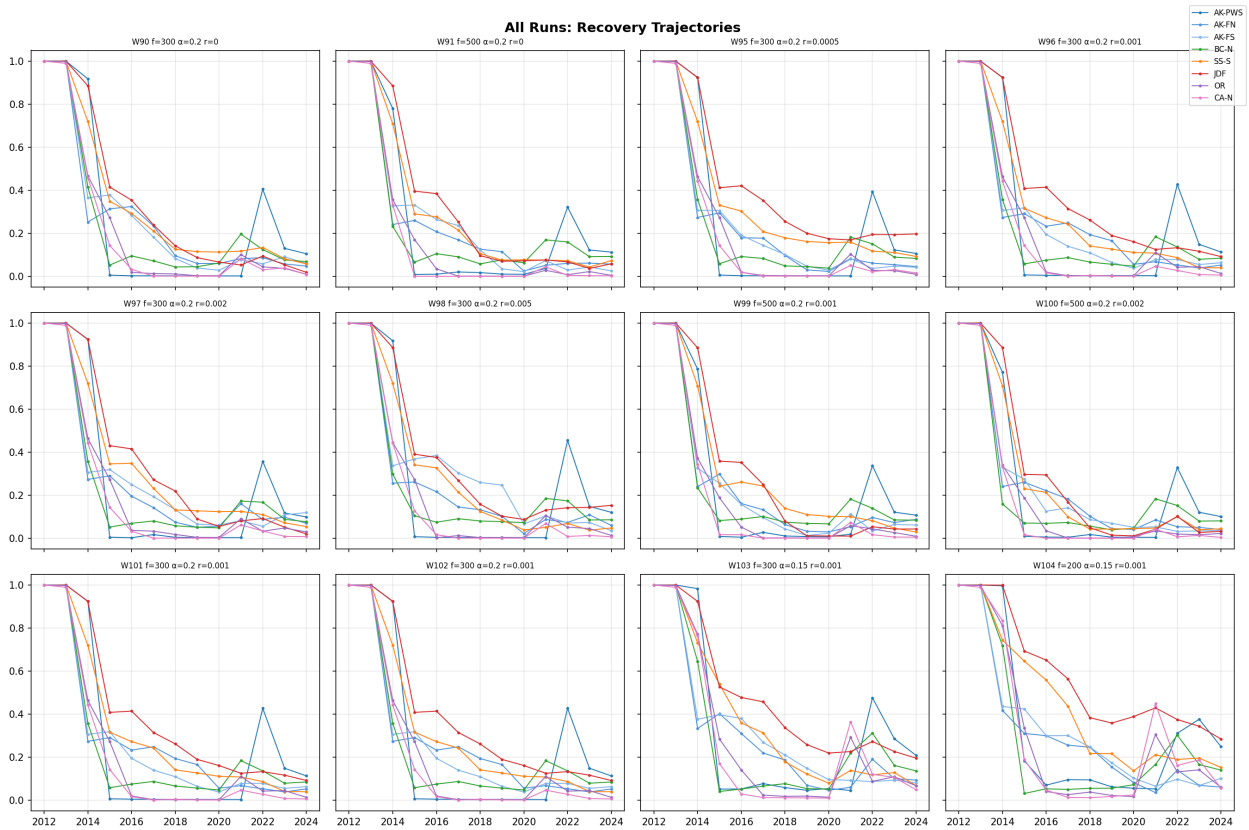

Figure 2: All 12 runs (2 references + 10 adaptation runs).

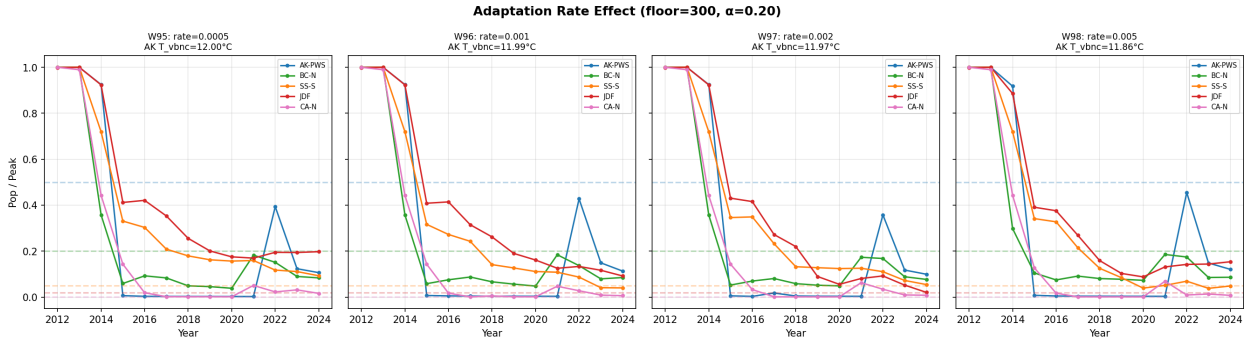

Figure 3: Increasing adaptation rate (W95–W98). Note final T_{vbnc} values barely change.

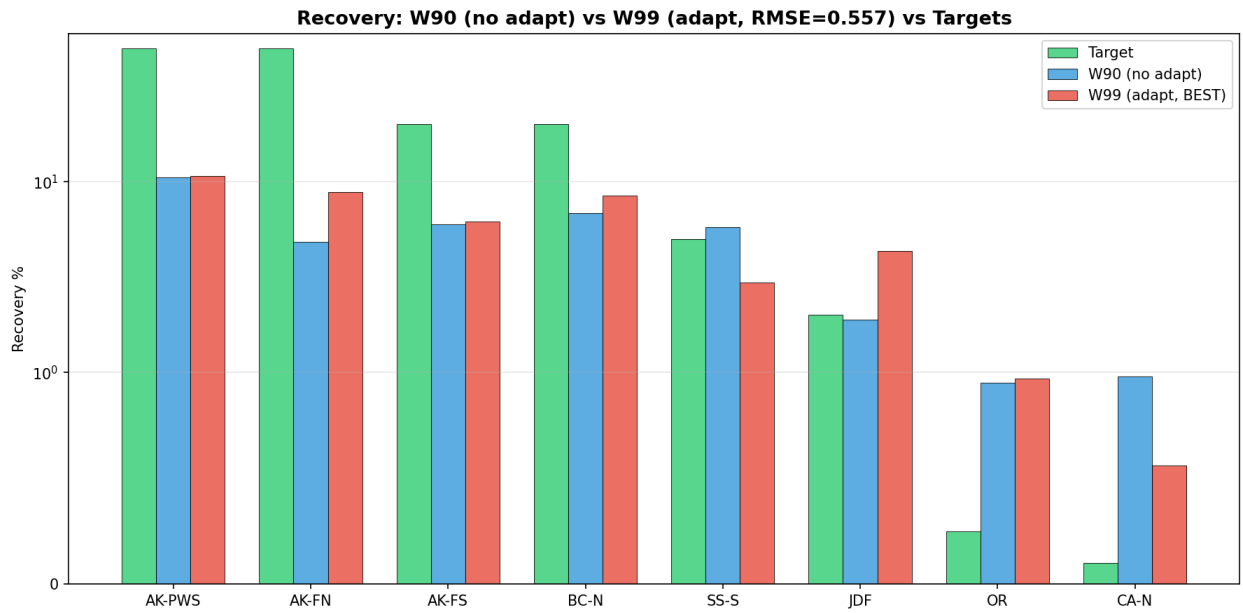

Figure 4: W90 vs W99 vs targets. AK-PWS remains ∼5× below target.

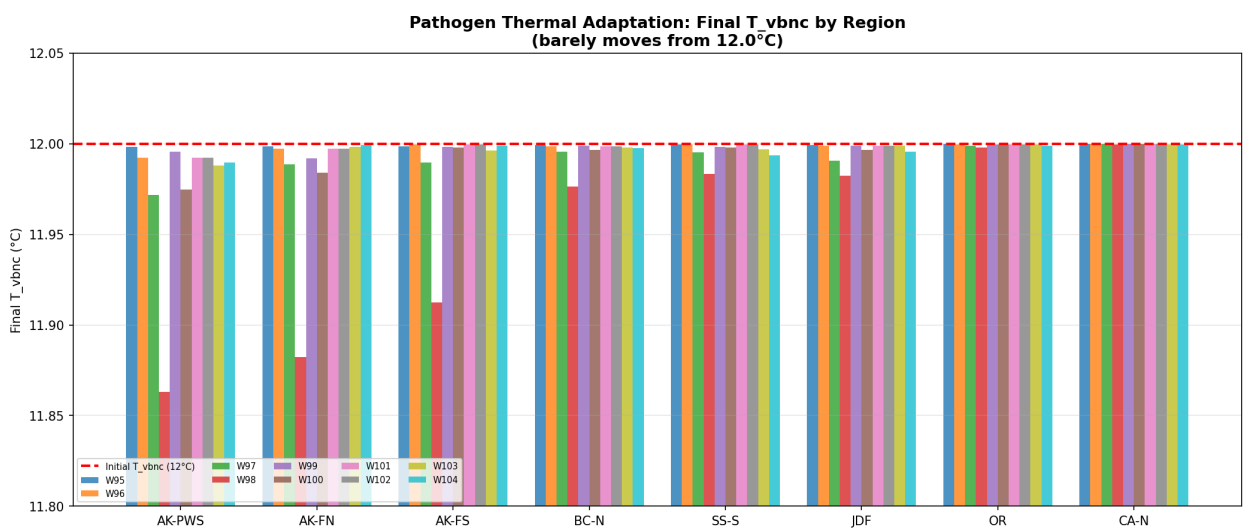

Figure 5: **The key negative result**: Final T_{vbnc} per region across all adaptation runs. All values cluster within 0.15°C of the initial 12.0°C. The mechanism is inert at these rates.

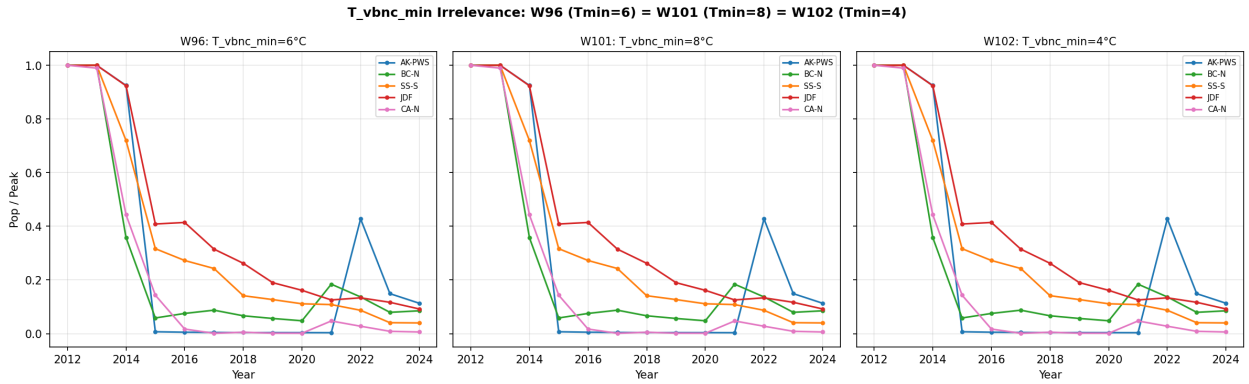

Figure 6: W96/W101/W102 produce *identical* results despite $T_{\text{vbnc,min}} = 6/8/4°\text{C}$. The floor is never approached.

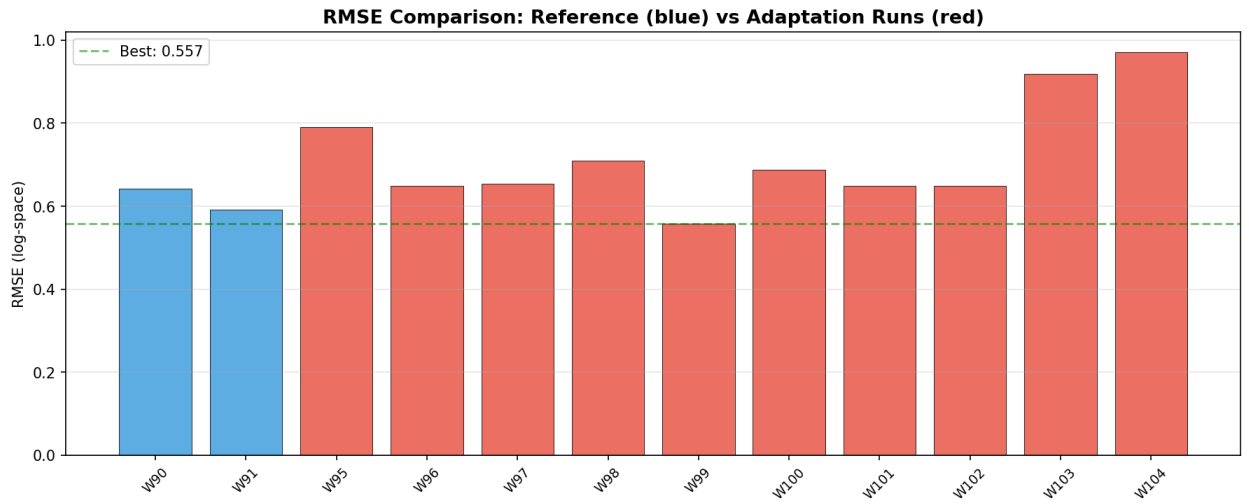

Figure 7: RMSE across all runs. W99 (0.557) is best, but improvement over W91 (0.591) is modest and not from adaptation.

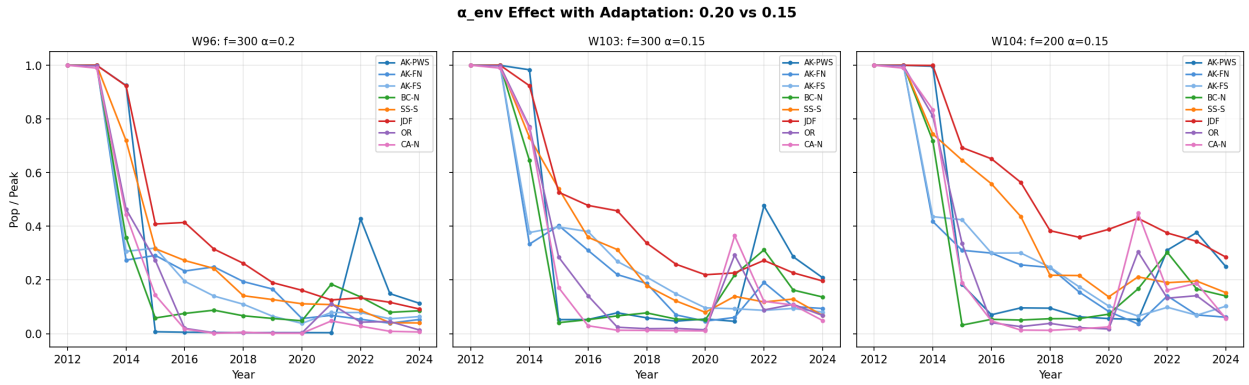

Figure 8: $\alpha = 0.15$ (W103/W104) vs $\alpha = 0.20$ (W96). Lower α reduces gradient, inflates southern recovery.